THE KERNEL OF THE MAP TO THE F-ADIC COMPLETION OF A DERIVED COMPLETE MODULE HAS SQUARE-ZERO INTERSECTION WITH THE IDEAL OF DIVISIBLE ELEMENTS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

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Editorial note. The counted statement span defines J as the intersection of the ideals f^nA and M' as the kernel of M -> lim M/f^nM; both definitions are delivered verbatim. The single reference inside the counted proof span is recorded as a non-load-bearing navigation pointer.

Allowed standard prerequisites (editorial).

- There are no maps from A_f to a module derived complete with respect to f. invoked at proof lines 27080-27081. Forces the constructed localization map to vanish, giving y_n = 0.
- The definition of derived completeness with respect to f and the module M' as the kernel of the map to the f-adic completion. used at statement lines 27068-27070. Fixes the objects and the meaning of the conclusion JM' = 0.
- Divisibility of elements of M' by arbitrary powers of f. used at proof line 27075. Produces the sequences x_n and hence the elements y_n.

Source reference. Original lemma 25850: sources/stacks-more-algebra/pinned-original-source.tex, source lines 25850–25850. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Lemma 0.2. *Let $f \in A$ be an element of a ring. Set $J = \bigcap f^n A$. Let M be an A-module derived complete with respect to f. Then $JM' = 0$ where $M' = \mathrm{Ker}(M \to \lim M/f^n M)$. In particular, if A is derived complete then J is an ideal of square zero.*

Proof. Take $x \in M'$ and $g \in J$. For every $n \geq 1$ we may write $x = f^n x_n$. Since g is in $f^n A$ we see that the element $y_n = g x_n$ in M' is independent of the choice of x_n. In particular, we may take $x_n = f x_{n+1}$ and we find that $y_n = f y_{n+1}$. Thus we obtain a map $A_f \to M$ sending $1/f^n$ to y_n. This map has to be zero as M is derived

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complete (Lemma 25850) and hence $y_n = 0$ for all n. Since $gx = gfx_1 = fy_1$ this completes the proof. $\square$